

# WHAT IS A MOLE?

The *mole* is a convenient unit to measure the very large number of particles present in a substance.

We count shoes in pairs: 2



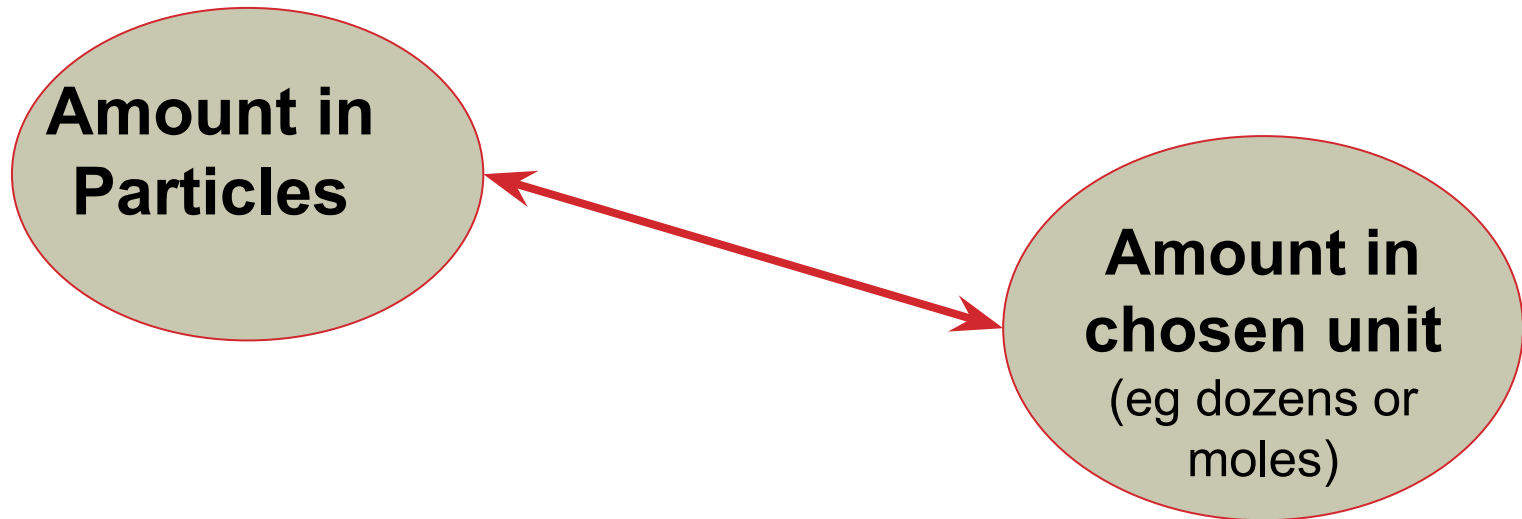
We count eggs by the dozen: 12



In chemistry we count particles in moles:  $6.02 \times 10^{23}$



# **CALCULATING FROM PARTICLES - Groups of 3 at boards**



**If you have 48 eggs  
how many dozens do  
you have?**

**Set up an EQUATION to get from particles to amount  
in DOZENS and solve it using:**

**$p$  = Number of eggs you have (particles of egg)**

**$N_D$  = number in 1 dozen**

**$n$  = the number of dozens you have**

**Using your equation,  
solve for the number  
of dozens in 103 pens.**

**Set up an EQUATION and solve it using:**

**$p$  = Number of eggs you have (particles of egg)**

**$N_D$  = number in 1 dozen**

**$n$  = the number of dozens you have**

**Now solve for how many pens are in 6.2 dozen.**

**DO NOT ERASE YOUR BOARD YET**

# WHAT DO WE MEAN BY PARTICLES?

| Substance                                                                   | Particles are called |
|-----------------------------------------------------------------------------|----------------------|
| Monoatomic elements<br>i.e. copper (Cu)                                     | atoms                |
| Diatomic elements (i.e. $O_2$ ) or<br>molecular compounds (i.e.<br>$H_2O$ ) | molecules            |
| Ionic compounds (i.e. NaCl)                                                 | formula units        |

# WHY CREATE A NEW UNIT?

Matter consists of an incredibly large number of atoms.

How many atoms does a 1 cm x 1 cm x 1 mm piece of copper (Cu) contain?

*Approximately 5,100,000,000,000,000,000,000 atoms!!!!!!*

or

$5.1 \times 10^{21}$  atoms!

This is very inconvenient!

PRESENTED BY



7

Chemistry

Introduction to the Mole



# DEFINING THE MOLE

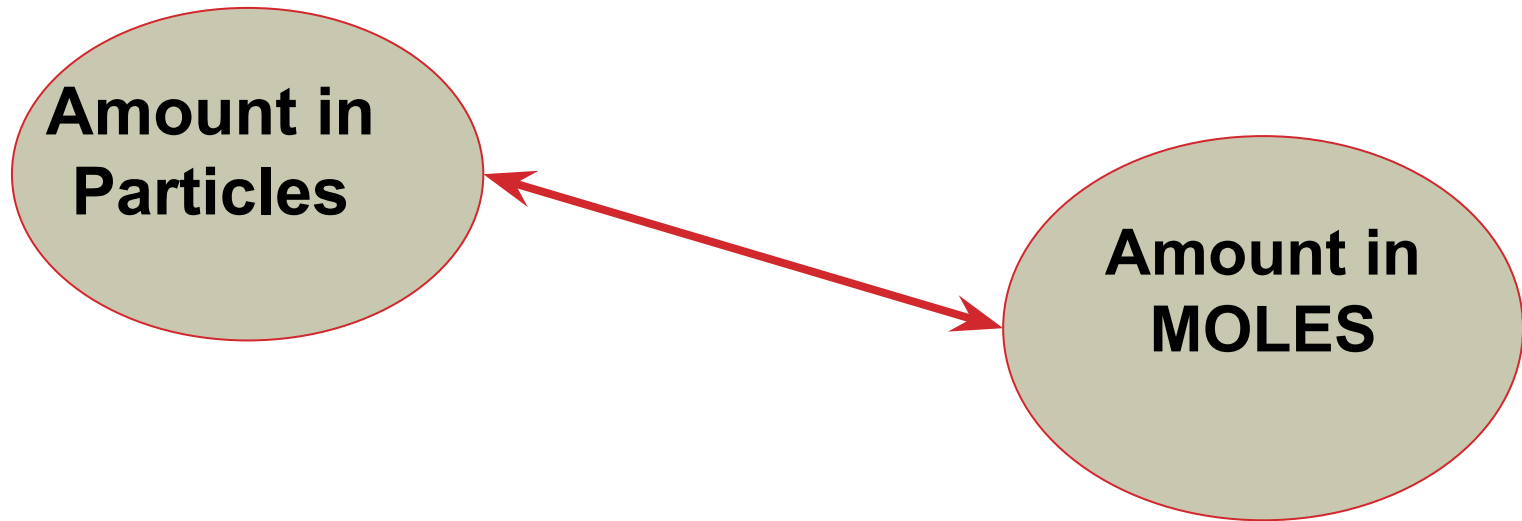
The mole was based on the number of atoms in 12 grams of the isotope carbon-12:

$$1 \text{ mole} = \frac{12.0 \text{ g}}{1.994 \times 10^{-24} \text{ g / atom}} = 6.02 \times 10^{23} \text{ atoms}$$

**This is known as Avogadro's Constant, or Avogadro's Number ( $N_A$ ).**



# **CALCULATING MOLES FROM PARTICLES - Return to Groups of 3 at boards**



**If you have  $6.02 \times 10^{23}$  eggs  
how many moles do you  
have?**

**Set up an EQUATION to get from particles to amount  
in MOLES and solve it using:**

**p = Number of particles you have**

**$N_A$  = number in 1 mole (this is Avogadro's number)**

**n = the number of moles you have**

**If you have  $1.81 \times 10^{24}$  eggs how many moles do you have?**

**Would your answer change if it was  $1.81 \times 10^{24}$  particles of sodium (Na)?**

**How about if it was  $1.81 \times 10^{24}$  particles of  $\text{CO}_2$ ?**

# CALCULATING PARTICLES FROM MOLES

How many  $\text{MgCl}_2$  formula units {ie particles} are present in 2.63 moles?

## REARRANGE YOUR EQUATION

$$\text{number of particles} = N = n \times N_A$$

$$N = (2.63 \text{ mol})(6.02 \times 10^{23} \text{ formula\_units / mol}) = 1.58 \times 10^{24} \text{ formula units}$$

**If you have 2 moles of Li,  
how many atoms  
(particles) do you have?**

**Again, would your  
answer change if it was  
2 moles of  $\text{Li}_2\text{O}$ ? 2 moles  
of  $\text{C}_2\text{H}_4$ ? Of Elephants?**

# CALCULATING THE MOLES FROM PARTICLES

The number of moles of a substance can be determined using this formula:

$$n = \frac{N}{N_A}$$

Where:  $n$  = number of moles.

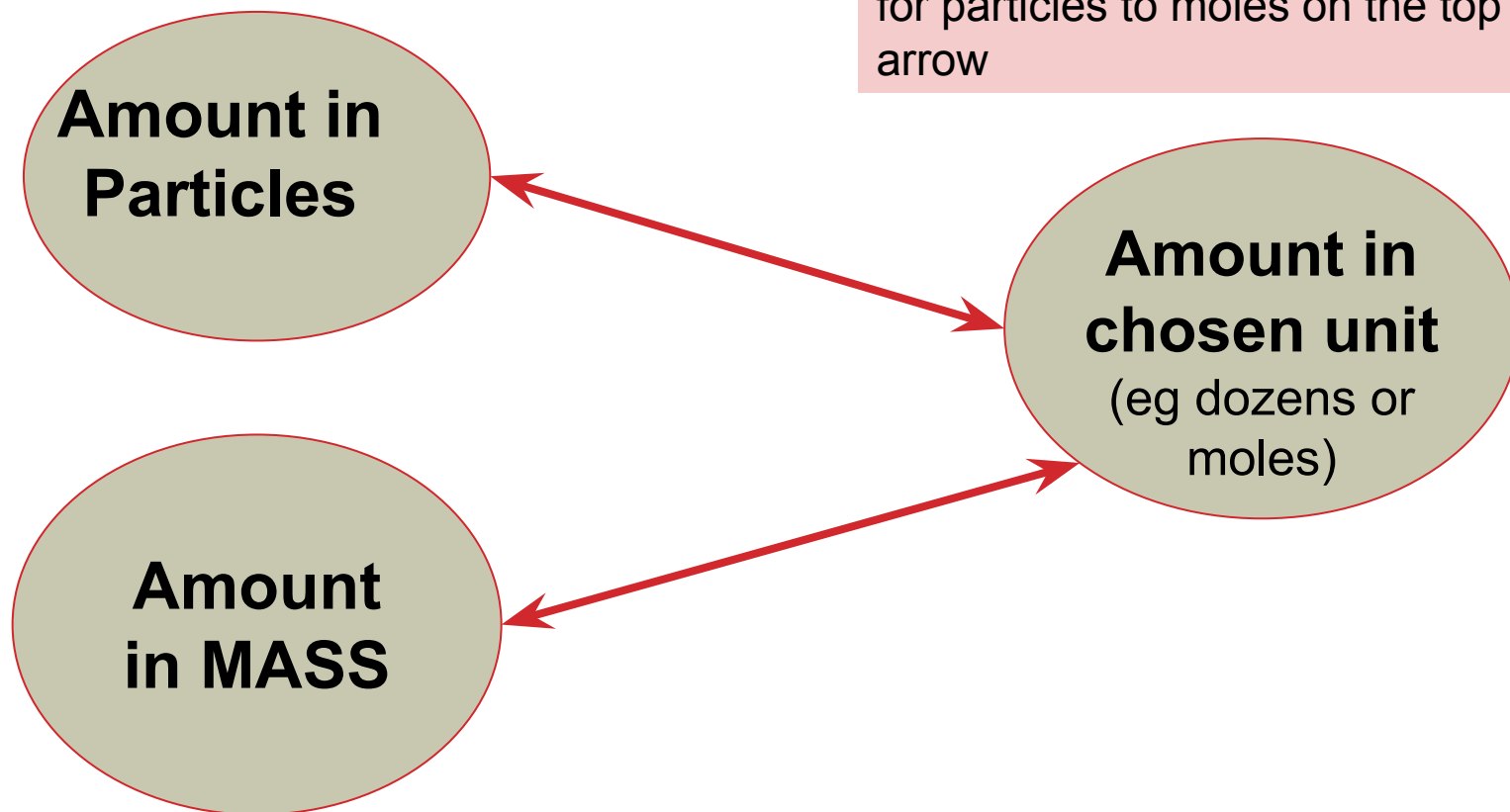
$N$  = number of particles. (a “p” may be used instead of N)

$N_A$  = Avogadro’s Constant

=  $6.02 \times 10^{23}$

# **CALCULATING FROM MASS - Groups of 3 at boards**

Make sure you have the equation  
for particles to moles on the top  
arrow



**If you have 60 g of eggs how many dozens do you have?  
(1 dozen eggs = 30g)**

**Set up an EQUATION to get from amount in mass to amount in DOZENS and solve it using:**

**m = mass of eggs you have**

**MM = mass of 1 dozen eggs**

**n = the number of dozens you have**



**Using your equation:  
If you have 3.2 dozen eggs  
how many grams do you  
have? (1 dozen eggs = 30g)**

**Using your equation:  
If you have 3.2 dozen eggs  
how many EGGS (particles)  
do you have? Solve on that  
arrow!**

**For your 3.2 dozen eggs converted to mass of eggs or number of eggs, what is true about the actual quantity of eggs?**

# CALCULATING THE MOLES FROM MASS

The number of moles of a substance can be determined using this formula:

$$n = \frac{m}{MM}$$

Where:  $n$  = number of moles.

$m$  = mass of sample

$MM$  = molar mass of the substance

We will look at determining molar mass in the next slides!

# **BAD MOLE JOKES**

**Q: How did Avogadro write to his friends?**

**A: By e-mole!**

**Q: Why was Avogadro so rich?**

**A: He's a multi-mole-ionare!**

**Q: What does Avogadro put in his hot chocolate?**

**A: Marsh-mole-ows!**

**Q: Why did Avogadro like Cindy Crawford?**

**A: She's his favorite super-mole-dle (and she has a mole).**

**Q: What did the generous Avogadro say when his friends crashed his party?**

**A: The mole the merrier!**